

- Q1.** What is the relation between the number of miles hiked x and the amount of water consumed y if $y = 2x + 1$?
- (A) One-to-one function
(B) Many-to-one relation
(C) One-to-many relation
(D) Many-to-many relation
(E) Neither relation nor function
- Q2.** If $f(x) = x^2 - 3$, what is $f(-2)$?
- (A) 1
(B) -1
(C) 5
(D) -5
(E) 7
- Q3.** A park ranger tracks the population y of deer over x years. If the data is modeled by $y = 5x + 20$, what is the range of y if x ranges from 0 to 10?
- (A) $20 \leq y \leq 70$
(B) $0 \leq y \leq 50$
(C) $20 \leq y \leq 50$
(D) $50 \leq y \leq 100$
(E) $70 \leq y \leq 100$
- Q4.** Determine the domain of $f(x) = \frac{1}{x-2}$.
- (A) $(-\infty, 2) \cup (2, \infty)$
(B) $(-\infty, 2]$
(C) $[2, \infty)$
(D) $(-2, 2)$
(E) $(-2, 2)$
- Q5.** A mapping f takes an input x to an output y such that $y = x^2$. Is f a function if the ordered pairs are $(-1, 1), (0, 0), (1, 1), (-1, -1)$?
- (A) Yes
(B) No
(C) Cannot be determined
(D) Only if $x \geq 0$
(E) Only if $x < 0$
- Q6.** For the function $y = 3x - 1$, which relation is equivalent?
- (A) $x = \frac{y+1}{3}$
(B) $x = \frac{y-1}{3}$
(C) $y = \frac{x+1}{3}$
(D) $y = \frac{x-1}{3}$
(E) $y = x - 1$
- Q7.** The fuel consumption y of a car is modeled by $y = 0.05x^2 + 2x + 10$ where x is the distance traveled in miles. What is the domain of this function if x must be non-negative?
- (A) $[0, \infty)$
(B) $(-\infty, 0]$
(C) $[0, 100]$
(D) $(-100, 100)$
(E) $(-\infty, -100)$
- Q8.** The vertical line test shows that $y = |x|$ is not a function. True or False?
- (A) True
(B) False
(C) Only in the first quadrant
(D) Only in the second quadrant
(E) Cannot be determined

Open Ended Questions (FRQ)

- Q9.** The population y of rabbits in a forest after x months is given by $y = 2x^2 + 5x + 10$. Find the range of y when $0 \leq x \leq 5$. Show your work and explain your reasoning.
- Q10.** Sketch the graph of $f(x) = \sqrt{x-2}$ and label the domain and range. Then, evaluate $f(6)$ and $f(10)$. Explain any restrictions on the domain.

Chef's Tips

- Chef's Tips: For multiple-choice questions, ensure students show work for partial credit.
- For free-response questions, encourage students to use proper notation and explain their reasoning.
- Remind students to check their units and ensure all answers are within the context of the given scenario.